

GRADES 7-10



EMERGENCY SUB PLANS PACK

for DIGITAL TECHNOLOGIES

★ Keep learning on track—even when plans change. ★



**NO-PREP
ACTIVITIES**



**READY-TO-USE
RESOURCES**



**PERFECT FOR
EMERGENCY
LESSONS**



**GREAT FOR
SUBSTITUTE
TEACHERS**



**ENGAGING
DIGITAL TECH
TASKS**



**PRINT OR
DIGITAL USE**



YOUR **LIFESAVER** IN THE CLASSROOM

 SAVE TIME

 REDUCE STRESS

 KEEP STUDENTS



MADE BY TEACHERS
FOR TEACHERS

Set this up while you are well

15 MINUTES

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a day when you are not.

The one job you have today

Print pages 3–6 of this pack and leave them in a clearly labeled folder where your substitute teacher will find them — top drawer, on the desk, or with the front office. Everything a substitute needs is in those four pages.

You do not need to print the lessons in advance. The substitute teacher chooses one on the day and photocopies it.

SETTING IT UP · FIFTEEN MINUTES, ONCE

- 1 **Print the whole pack once.** Put it in a display folder or bind it. Write your name and room number on the cover.
- 2 **Print pages 3–6 a second time** and clip them to the front. These are the pages a substitute teacher reads first.
- 3 **Tell your department head where it lives.** On a genuine emergency day you will not be answering your phone, so somebody else needs to know.
- 4 **Cross off the lessons you have already used.** There is a tracker on page 4. Two years of substitute days will not repeat if you keep it current.
- 5 **Optional: leave a class list** and a note about any students who need specific support, seating or a reader.

Why these lessons are unplugged

On a substitute day the laptop cart is often locked, the login server is down, or nobody knows the room password. Every lesson here runs on paper.

Each lesson also has an **If the computers are working** note, so the same sheet still works when technology cooperates.

Why they are not busy work

Every lesson targets a genuine Digital Technologies concept and produces marked evidence you can put in a folder.

If you are away for a week, running four of these in sequence is a defensible unit of work, not a holding pattern.

A word about the answer keys

Every answer key sits on the **last page of its lesson**. If you hand a lesson to a substitute teacher, hand them the answer key too — it is the single thing that lets a non-specialist mark with confidence and answer a student who says *“but why?”*

This is Volume Three

Volume 3 stands completely on its own and shares nothing with Volumes 1 or 2. There is no binary conversion, no flowchart, no bubble sort, no data cleaning and no compression here. Owning all three gives you thirty substitute lessons that never repeat.

You do not need to know this subject

You have walked into a Digital Technologies class. You may have never taught the subject. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to know how to code. Every lesson is a paper worksheet with a full answer key. Your job is to hand it out, keep the room settled, and check work as students finish. You are not expected to teach the content.

DO THESE FOUR THINGS

- 1 **Pick one lesson** from the table on the next page. Any of them works for any class from Grade 7 to Grade 10. If you cannot decide, use Lesson 03.
- 2 **Photocopy the student pages only.** Each lesson tells you exactly which page numbers to copy. One copy per student.
- 3 **Keep the teacher page and the answer key.** Do not photocopy those.
- 4 **Read the teacher page once** before the bell. It takes about two minutes and includes the exact words you can say to start the lesson.

If a student says “we have done this”

Ask them to prove it by completing the extension section at the end. If they genuinely have, switch them to a different lesson from the table — the pack has ten, and they will not have done them all.

If you finish early

Every lesson has an extension task on its final student page. Point students there. There is also a discussion prompt on the teacher page you can run as a whole class for the last five minutes.

What to leave for the regular teacher

Collect the completed sheets. Fill in the handover note on the last page of this pack — it takes ninety seconds and tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	Lists, Indexes and Off-by-One	Data structures and indexing	★★	55	7-9
02	True, False and In Between	Boolean logic and conditions	★★	55	12-14
03	Inside the Box	Hardware and how a CPU works	★	55	17-19
04	Where Did the File Go?	File systems, paths and backups	★	55	22-24
05	Is It Finished?	Requirements, evidence and evaluation	★★	60	27-29
06	Teaching a Machine	Machine learning, training data and bias	★★	60	32-34
07	The Internet's Address Book	DNS, addresses and protocols	★★	55	37-39
08	How a Search Engine Ranks	Indexing, ranking and relevance	★★★	60	42-44
09	Machines That Sense and Act	Control algorithms and thresholds	★★	55	47-49
10	Whose Work Is It?	Copyright, licensing and attribution	★	55	52-54

Easiest to supervise

03, 04 and 10. Reading, judgment and argument, with a definite answer waiting at the end of each one.

Best for a quiet, working class

01, 02 and 08. Puzzle-like with exact answers. Strong students will race; that is what the extensions are for.

Best for a restless class

06, 07 and 09. Counting patterns, taking an address apart and steering a robot through a maze. Noisier, but productively so.

Used-lesson tracker · for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Lists and indexes			
02 — Boolean logic			
03 — Hardware			
04 — Files and backups			
05 — Requirements			
06 — Machine learning			
07 — DNS and addresses			
08 — Search ranking			
09 — Robots and control			
10 — Copyright and licenses			

The first ten minutes

SCRIPT

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

"Come in, sit down, and leave your bags off the desks. You will not need a computer today. Everything is on paper."

THEN, ONCE THEY ARE SEATED

"Your teacher is away. My name is _____. We are doing a proper Digital Technologies lesson today, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their regular teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Settle, hand out sheets	Use the script above
5-10	Read the opening page together	Ask one student to read it aloud
10-45	Students work through the tasks	Circulate. Check answers against the key
45-55	Extension task for those who finish	Point them to the last student page
55-60	Collect sheets, pack up	Fill in the handover note

If a student refuses to work

Offer the smaller ask: *"Just do question one and show me."* Most refusals are about not knowing where to start, not defiance.

Note it on the handover sheet. Do not escalate a single lesson into a battle you will not be there to finish.

If the whole class claims they cannot do it

Do the first question together on the board. Every lesson has a worked example on the first student page.

If it is genuinely too hard for the grade level, switch to Lesson 03 or Lesson 10 — both need no prior knowledge at all.

Photocopying reminder

Student pages only. Each lesson's teacher page states the exact page range. The **answer key is always the last page** of the lesson — keep it.

Lists, Indexes and Off-by-One

2 MINUTES

Students look up items in a list by index, trace four operations that change it, then hunt down the single most common bug in programming: the loop that runs one time too many or too few.

You do not need to know this subject

Every answer is one word or one short list, and every list is written out after each step in the key. There is no code to write.

Photocopy these pages only

Pages 7-9

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Crossing out is better than rubbing out, so the working stays visible.

WHAT TO SAY TO START

"A list is a row of numbered boxes, and the numbering starts at zero, not one. Almost every mistake in this lesson comes from that one sentence, and so does almost every mistake made by professionals."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-12	Read the panel and the worked example	Say the word index out loud repeatedly
12-24	Task 1.1 — look things up	Every answer is definite. Grade as you circulate
24-36	Task 1.2 — trace the operations	Key has the list after every step
36-48	Task 1.3 — the off-by-one hunt	Four bugs, four fixes
48-55	Tasks 1.4 and 1.5, collect sheets	Discussion prompt below

Questions students will ask

"Why does it start at zero?"

The index is really a distance from the start of the list. The first item is zero steps away. Task 1.5 asks them to work that out.

"What does the minus one mean?"

Count backwards from the end. [-1] is the last item, whatever the length.

"Is index 8 of an 8-item list really an error?"

Yes. The valid indexes are 0 to 7. This trips up everybody once and then never again.

If they finish early

Task 1.4 moves to a two-dimensional list, which is how every seating plan, spreadsheet and game board is stored. Strong students can then work out how many indexes a chessboard needs and write the one for the bottom-left square.

If the computers are working

Students can type a list into any Python environment and try the index that should fail. The error message names the exact problem, which is unusually helpful for a computer.

Curriculum focus · for the regular teacher's records

Data structures: lists and arrays; indexing and iteration; identifying and correcting logic errors in loops. Discussion prompt for the last five minutes: *"Counting from zero causes thousands of bugs a year. So why has no language ever fixed it?"*

Numbered boxes

14 MIN

NAME

CLASS

DATE

A list is a row of boxes with numbers on them

The number is the **index**. The first box is index **0**, not 1. A list of five items has indexes 0, 1, 2, 3 and 4 — and asking for index 5 is an error, not an empty box.

worked.py

PYTHON

```
names = ["Ari", "Bec", "Cam", "Dev", "Eli"]

print(names[0]) # Ari
print(names[3]) # Dev
print(len(names)) # 5
print(names[-1]) # Eli
print(names[5]) # error
```

What to notice

len gives the number of items, so the last valid index is always **len – 1**.

[-1] counts backwards from the end and always finds the last item.

The final line does not print a blank. It stops the program.

Task 1.1 · Look it up

DO

subjects.py

PYTHON

```
subjects = ["Math", "English", "Science", "Art",
            "PE", "Music", "History", "Digital Tech"]
```

#	QUESTION	ANSWER
a	subjects[0]	
b	subjects[4]	
c	len(subjects)	
d	The index of "Music"	
e	subjects[-1]	
f	subjects[8]	
g	The highest index that works	

h. Write the rule connecting **len** to the highest valid index: _____

Changing the list

18 MIN

Task 1.2 · Trace the operations

TRACE

The list starts as `[4, 9, 2]`. Carry out each instruction on the result of the one before it. Write the whole list every time.

STEP	INSTRUCTION	THE LIST IS NOW	LENGTH
1	Add 7 to the end		
2	Insert 5 at index 1		
3	Remove the item at index 0		
4	Replace index 2 with 8		
5	Add a copy of the item at index 0 to the end		

- Which step changed the index of every item after it? _____
- Why is that dangerous inside a loop? _____

Task 1.3 · The off-by-one hunt

DEBUG

The list `scores` holds **six** numbers. Each loop below is wrong by exactly one. Say what goes wrong, then fix it.

#	THE LOOP	WHAT GOES WRONG	THE FIX
a	<code>for i in range(5):</code>		
b	<code>for i in range(1, 6):</code>		
c	<code>for i in range(7):</code>		
d	<code>while i <= 6:</code>		

Two of these four crash and two do not. Which two are more dangerous, and why?

Task 1.4 · The seating plan

EXTENSION

A classroom is stored as a list of rows, and each row is a list of names. Rows and seats both start at zero.

ROW	SEAT 0	SEAT 1	SEAT 2	SEAT 3
0	Ari	Bec	Cam	Dev
1	Eli	Fen	Gus	Hal
2	Ida	Jem	Kai	Lou

#	QUESTION	ANSWER
a	<code>seats[1][2]</code> is	
b	Kai's position is <code>seats[][]</code>	
c	How many seats in total?	
d	Why is <code>seats[3][0]</code> an error?	

e. Write the two indexes you would use to reach the back right corner of a room with 8 rows of 6:

Task 1.5 · The last word

THINK

The index is not really a position. It is a **distance from the start**. Use that sentence to explain, in one or two sentences, why the first item is number zero.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials

Task 1.1 · looking things up

a. Math · **b.** PE · **c.** 8 · **d.** 5 · **e.** Digital Tech · **f.** An error — the program stops. `IndexError: list index out of range` · **g.** 7.

h. The highest valid index is always `len - 1`. Any student who writes that sentence has the whole lesson.

Watch for students answering **f** with ‘nothing’ or ‘blank’. The list does not contain an empty box at index 8; there is no box at all.

Task 1.2 · the trace

STEP	THE LIST	LENGTH
Start	[4, 9, 2]	3
1	[4, 9, 2, 7]	4
2	[4, 5, 9, 2, 7]	5
3	[5, 9, 2, 7]	4
4	[5, 9, 8, 7]	4
5	[5, 9, 8, 7, 5]	5

a. Steps 2 and 3 both did — inserting and removing shift everything after them.

b. A loop that has already decided it will run six times does not notice that item 4 is now item 3. Removing items from a list while looping over it skips items, and it is one of the most confusing bugs a beginner meets.

Task 1.3 · the four off-by-one bugs

#	WHAT GOES WRONG	THE FIX
a	Runs for indexes 0–4 only. The sixth mark is silently ignored	<code>range(6)</code> or <code>range(len(scores))</code>
b	Starts at index 1, so the first mark is skipped	<code>range(0, 6)</code> , or simply <code>range(6)</code>
c	Reaches index 6, which does not exist. Crashes on the last pass	<code>range(6)</code>
d	Same fault: <code>i</code> reaches 6 and the lookup fails	<code>while i < 6:</code>

a and b are the dangerous ones. c and d crash, which tells you immediately that something is wrong. a and b run perfectly and quietly produce an average of the wrong five scores — a number nobody will ever question.

`range(len(scores))` is the answer worth teaching: it is correct whatever the list length becomes later.

Task 1.4 · the seating plan

a. Gus · **b.** `seats[2][2]` · **c.** 12 · **d.** There are only rows 0, 1 and 2. Row 3 does not exist.

e. `seats[7][5]` . Almost every class writes `[8][6]` first — the same off-by-one, arriving in a new costume.

The first index is the row, the second is the seat. Reading it as ‘the third item of the second row’ is the habit worth building.

Task 1.5 · the closing question

The list is one continuous block of memory. The index tells the machine how far to move from the start to reach the item you want — and the first item requires no movement at all, so it is zero.

Accept any answer reaching ‘it counts the steps from the beginning, not the items’.

Worth adding: languages that start at one have to subtract one every single time an item is fetched. The bug never disappears; it just moves from the programmer into the machine.

True, False and In Between

2 MINUTES

Students complete truth tables, translate six real rules into AND, OR and NOT, then meet a condition that can never be true and work out why no computer will ever warn them about it.

You do not need to know this subject

Every answer is TRUE or FALSE and every table is completed in the key. If you can decide whether two things are both true, you can grade this lesson.

Photocopy these pages only

Pages 12-14

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

"Computers have exactly two words for the state of the world, and three ways to combine them. Everything a machine has ever decided was built out of what is on this sheet."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-12	Read the panel and the worked table	Do the AND table on the board
12-24	Task 2.1 — the truth tables	Every cell is in the key
24-36	Task 2.2 — rules into logic	Two of the six are broken. Do not say which
36-48	Task 2.3 and 2.4 — the alarm, then the discovery	Key has both tables
48-55	Task 2.5, collect sheets	Discussion prompt below

Questions students will ask

"Does OR mean one or the other, or both?"

In computing, OR is true if either is true and also if both are. Everyday English usually means the other thing, which is why this confuses people.

"Is NOT the same as false?"

No. NOT flips whatever it is given. NOT false is true.

"How can a condition be impossible?"

That is Task 2.2d and it is the best question on the sheet. Let them find it.

If they finish early

Task 2.4 has students discover a rule that professionals use daily without knowing its name. Strong students can then rewrite the alarm condition in Task 2.3 using only NOT and OR, which is genuinely how some hardware is built.

If the computers are working

Any logic gate simulator will let students build the alarm circuit and toggle the inputs. Insist the paper truth table is completed first, or the simulator does the thinking.

Curriculum focus · for the regular teacher's records

Boolean logic and truth tables; branching conditions; combining conditions with AND, OR and NOT; identifying logic errors. Discussion prompt for the last five minutes: *"A condition that can never be true is perfectly legal code. Why can the computer not simply tell you?"*

Two words and three operations

14 MIN

NAME _____

CLASS _____

DATE _____

AND

True only when **both** sides are true. Everything else is false.

OR

True when **either** side is true, including when both are.

NOT

Flips it. NOT true is false; NOT false is true.

Worked example · A AND B

A	B	A AND B
F	F	F
F	T	F
T	F	F
T	T	T

Four rows, because two things that can each be true or false make exactly four situations.

Task 2.1 · Complete the tables

DO

A	B	A OR B
F	F	
F	T	
T	F	
T	T	

A	B	A AND NOT B
F	F	
F	T	
T	F	
T	T	

How many rows would a table with **three** inputs need? _____ Why? _____

Task 2.2 · Rules into logic

TRANSLATE

Write each rule using AND, OR and NOT. Two of these six are written wrongly already — find them and say what actually happens.

#	THE RULE AS SOMEBODY WROTE IT	AS LOGIC, OR WHAT IS WRONG WITH IT
a	Free entry if you are under 5 or 65 and over	
b	The alarm sounds if the door is open and the system is armed	
c	if day == "Sat" or "Sun":	
d	if age > 12 and age < 5:	
e	Show the login screen if the user is not signed in	
f	Discount for students or seniors, but not at peak times	

Task 2.3 · The shop alarm

TRACE

The alarm sounds when: **the system is armed AND a door is open AND NOT a valid key was used.**
Work out what happens in each situation.

#	ARMED?	DOOR OPEN?	VALID KEY?	ALARM SOUNDS?
1	T	T	F	
2	T	T	T	
3	T	F	F	
4	F	T	F	
5	F	F	T	

- a. A thief props the door open before the shop is armed for the night. Which row is that, and what does the alarm do? _____
- b. Write one extra condition that would close that hole: _____

Task 2.4 · Find the rule nobody told you

DISCOVER

Fill in both middle columns, then compare the last two.

A	B	A AND B	NOT (A AND B)	(NOT A) OR (NOT B)
F	F			
F	T			
T	F			
T	T			

- a. What do you notice about the last two columns? _____
- b. Rewrite 'it is not both raining and cold' using OR: _____

Task 2.5 · Legal, and impossible

THINK

grades.py

PYTHON

```
if mark > 100 and mark < 0:
    print("Impossible mark")
```

```
if mark >= 50 or mark < 50:
    print("Every student sees this")
```

#	QUESTION	ANSWER
a	How often does the first message print?	
b	How often does the second one print?	
c	Is either of these a syntax error?	
d	Which one is more dangerous in a real program, and why?	

e. Explain in one sentence why a computer cannot simply refuse to accept a condition that is never true.

Task 2.6 · The last word

THINK

Every decision a computer has ever made was built from TRUE, FALSE, AND, OR and NOT. Write one sentence explaining why that is either impressive or worrying — your choice, but give a reason.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Task 2.1 · the tables

A OR B: F, T, T, T — only the first row is false.

A AND NOT B: F, F, T, F — true only when A is true and B is false.

Three inputs need 8 rows. Each extra input doubles the table: 2, 4, 8, 16. Students who answer 6 have added instead of doubled, which is worth stopping to correct.

Task 2.3 · the alarm

1 — yes. Armed, open, no key. This is the burglary the alarm exists for.

2 — no. A valid key was used; this is the owner.

3, 4, 5 — no.

a. Row 4. The alarm does nothing, because the system was not armed when the door was opened, and arming it later does not re-check a door that is already open.

b. Model answer: refuse to arm at all while any door is open, or add 'OR a door was already open at the moment of arming'. Accept any answer that checks the state at arming time. Real alarm panels do exactly this and it is why they beep at you.

Task 2.5 · legal and impossible

a. Never. No number satisfies both halves.

b. Always. Every number is either 50 or more, or less than 50.

c. No. Both are perfectly valid code. The rules of the language are obeyed; only the meaning is nonsense.

d. The first. Code that never runs usually means a check the programmer believed was protecting them — so the protection was never there. The second is merely pointless.

e. Deciding whether a condition can ever be true means predicting every possible value the program might produce, which in general cannot be done. Simple cases can be caught, and good tools warn about them; the general problem is impossible.

Task 2.2 · rules into logic

a. `age < 5 OR age >= 65`

b. `door_open AND armed`

c. Broken. The computer reads it as 'if day equals Sat' OR 'if the word Sun', and a non-empty word counts as true — so this is true *every day of the week*. Correct: `day == "Sat" or day == "Sun"`.

d. Broken. No number is both above 100 and below 0. The block never runs and the program never complains.

e. `NOT signed_in`

f. `(student OR senior) AND NOT peak` — the brackets are the whole answer. Without them the rule means something different.

Task 2.4 · what they discovered

A AND B: F, F, F, T. **NOT (A AND B):** T, T, T, F. **(NOT A) OR (NOT B):** T, T, T, F.

a. The last two columns are identical. NOT applied to an AND becomes an OR of the two NOTs.

b. 'It is not raining, or it is not cold'.

This is **De Morgan's law**, and it is worth naming once they have found it themselves. Programmers use it constantly to turn an awkward negative condition into a readable one.

Task 2.6 · the closing question

Both answers are defensible and the credit is for the reasoning.

Impressive: every calculation, image, message and model ever produced by a machine reduces to these five things repeated at enormous speed.

Worrying: a system that only knows true and false has no room for 'it depends', 'probably' or 'not enough information', so somebody has to force every messy human situation into one of two boxes before the machine can act on it.

Inside the Box

Students match ten components to their jobs, diagnose four slow computers, then run a five-instruction program by hand the way a processor does. Concrete, physical and easy to supervise.

You do not need to know this subject

Every answer is a component name or a number, and the key gives both. No technical background is needed to mark it.

Photocopy these pages only

Pages 17-19

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

"There are about six things inside every computer ever built, and once you know which one does what, you can work out why any computer is slow without opening it."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-14	Read the panel and the memory-versus-storage idea	The desk analogy does the work
14-26	Task 3.1 — match the components	Ten answers, all in the key
26-38	Task 3.2 — four slow computers	Expect argument about the last one
38-50	Task 3.3 — be the processor	Exact values. Key has every step
50-55	Tasks 3.4 and 3.5, collect sheets	Discussion prompt below

Questions students will ask

"What is the difference between memory and storage?"

Memory is the desk: fast, small, cleared every night. Storage is the filing cabinet: slow, huge, still there tomorrow.

"Is the GPU just for games?"

No. It does thousands of simple sums at once, which is why it is also used for video, and for training AI models.

"Why is my computer slow?"

The honest answer is that it depends which part is waiting. That is exactly Task 3.2.

If they finish early

Task 3.4 asks students to put six kinds of storage in speed order. Strong students can then work out why a processor spends most of its time waiting, and what a cache is for.

If the computers are working

Any task manager or system monitor shows processor, memory and disk usage live. Open something heavy and watch which number climbs — that is Task 3.2 happening in real time.

Curriculum focus · for the regular teacher's records

Digital systems: hardware components and their functions; the role of the processor, memory and storage; the fetch-execute cycle.

Discussion prompt for the last five minutes: *"Your phone has 8 GB of memory and 256 GB of storage. Why not simply have 256 GB of memory?"*

The desk and the filing cabinet

14 MIN

NAME

CLASS

DATE

One useful picture

Think of doing homework at a desk. The **processor** is you. **Memory** is the desk surface: everything you are working on right now, cleared completely when you leave. **Storage** is the filing cabinet: slower to reach, far bigger, and still full tomorrow morning.

Everything else in the box exists to move things between those three, or to keep them cool and powered.

Task 3.1 · Match the component to the job

CLASSIFY

COMPONENT	WHAT IT DOES	COMPONENT	WHAT IT DOES
CPU		Power supply	
RAM		Heatsink and fan	
SSD		Network card	
GPU		Cache	
Motherboard		Optical drive	

a. Which two of these ten lose everything the moment the power goes off? _____

b. Which one is the only part that does not process, store or move data at all? _____

Task 3.2 · Four slow computers

DIAGNOSE

#	THE SYMPTOM	WHICH PART IS THE BOTTLENECK	WHY
a	Fine with one program; crawls with twelve tabs open		
b	Takes four minutes to copy a folder of vacation photos		
c	Video looks fine but the game stutters badly		
d	One long calculation takes an hour, everything else is fine		

A shop offers to fix computer **a** by fitting a faster processor. Explain why that is the wrong repair.

Task 3.3 · Be the processor

TRACE

A very simple machine has one working register, called the **accumulator**, and a memory box called **A**. Run the program one instruction at a time.

tiny.asm

MACHINE

```
1 LOAD 6
2 ADD 4
3 STORE A
4 ADD 5
5 OUT
```

STEP	ACCUMULATOR	BOX A	OUTPUT
1			
2			
3			
4			
5			

a. Does **STORE** empty the accumulator? _____ **b.** What is printed? _____

c. The machine repeats three steps for every instruction: fetch it, work out what it means, do it. Which of the three would be fastest to skip, and what would break? _____

Task 3.4 · Order by speed

RANK

Number these 1 (fastest for the processor to reach) to 6 (slowest).

KIND OF STORAGE	1-6	KIND OF STORAGE	1-6
Hard disk with spinning platters		Solid state drive	
Cache inside the processor		Registers inside the processor	
Files on a cloud server		RAM	

a. As you move down your list, two things get better and one gets worse. What are they?

b. Why does a computer bother having *all six* instead of just the fastest one it can afford?

Task 3.5 · The last word

THINK

Your phone has 8 GB of memory and 256 GB of storage. Give two different reasons why it does not simply have 256 GB of memory instead.

- 1 _____
- 2 _____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

 Teacher initials _____

Task 3.1 · the components

- **CPU** — carries out instructions; the part that actually computes
- **RAM** — holds what is running right now; wiped at power off
- **SSD** — keeps files permanently, no moving parts
- **GPU** — does thousands of simple sums at once, for graphics and AI
- **Motherboard** — connects everything and carries data between parts
- **Power supply** — converts mains power to the low voltages the parts need
- **Heatsink and fan** — removes heat so the processor can keep running fast
- **Network card** — sends and receives data over cable or wifi
- **Cache** — a very small, very fast memory close to the processor
- **Optical drive** — reads discs; increasingly absent from new machines

- a. RAM and cache. Both are **volatile** — the contents exist only while powered.
- b. The heatsink and fan. Every other part handles data; cooling only makes that possible. Accept the power supply with reasoning — it is the same argument.

Task 3.2 · the four bottlenecks

- a. **RAM**. Too many programs at once means the machine shuffles data out to storage to make desk space, and storage is far slower.
- b. **Storage**. Copying is limited by how fast the drive can read and write, not by thinking speed.
- c. **GPU**. Video playback is handled easily; games ask for millions of new pixels every second.
- d. **CPU**. One long calculation cannot be shared out, so it is limited by the speed of a single core.
- The shop is wrong** because the processor in a is not the part that is struggling — it is idle, waiting. Fitting a faster one makes it wait faster. More memory is the fix, and it is cheaper.

Task 3.3 · the trace

STEP	ACCUMULATOR	BOX A	OUTPUT
1 LOAD 6	6	—	
2 ADD 4	10	—	
3 STORE A	10	10	
4 ADD 5	15	10	
5 OUT	15	10	15

- a. No. **STORE** copies the value out; the accumulator keeps it. Students who empty it get 5 at the end, which is the most common error here.
- b. 15.
- c. Skipping the fetch is impossible — without it there is no instruction. Skipping the decode is what happens in practice: processors predict and pre-decode ahead of time, and when the prediction is wrong the work is thrown away. Accept any thoughtful answer; the reasoning is the mark.

Task 3.4 · the hierarchy

- 1 registers · 2 cache · 3 RAM · 4 SSD · 5 hard disk · 6 cloud.
- a. Going down the list, capacity gets bigger and cost per gigabyte gets cheaper; speed gets worse. That trade is the entire reason the hierarchy exists.
- b. Registers cost thousands of times more per byte than storage, so a machine built entirely from the fastest kind would be unaffordable and tiny. Keeping the data you need in a moment close and the rest far away gives most of the speed for a fraction of the price.

Task 3.5 · the closing question

- Cost**. RAM is many times more expensive per gigabyte than flash storage. A phone with 256 GB of RAM would cost more than a car.
- Volatility**. RAM forgets everything when the power stops. A phone whose entire contents vanished on a flat battery would be useless.
- Accept power consumption as a third reason — RAM must be constantly refreshed to hold its contents, and 256 GB of it would flatten the battery in minutes.

Where Did the File Go?

2 MINUTES

Students read a folder tree and write full paths, work out what an extension really does, then decide which of four so-called backups would actually survive a laptop being stolen.

You do not need to know this subject

Everything here is reading a diagram and making a judgment. The key gives the exact path for every file and the reasoning for every backup.

Photocopy these pages only

Pages 22-24

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

"Everybody in this room has lost a file. Today you find out where files actually live, why renaming one does not change what is inside it, and why the copy you think is a backup probably is not."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-12	Read the tree together	Trace one path aloud with a finger
12-24	Task 4.1 — write the paths	Exact answers. Grade as you circulate
24-36	Task 4.2 — extensions	The rename question is the memorable one
36-48	Task 4.3 — is it really backed up	Expect argument about the sync folder
48-55	Tasks 4.4 and 4.5, collect sheets	Discussion prompt below

Questions students will ask

"Are folders real?"

Not physically. They are labels the file system keeps so humans can find things. The data itself is scattered across the drive.

"Can two files have the same name?"

Yes, in different folders. That is most of what folders are for.

"If I rename it back, is it fixed?"

Yes, and that is the point of Task 4.2. Nothing inside the file ever changed.

If they finish early

Task 4.4 asks what happens to a deleted file. Strong students can then explain why selling a phone without wiping it is dangerous even after everything has been deleted.

If the computers are working

Students can look at the address bar of any file manager and read the path they have just walked to. Turning on hidden file extensions is a two-minute demonstration worth doing.

Curriculum focus · for the regular teacher's records

File management in digital systems; naming, organizing and locating files; file types; data backup, recovery and secure deletion. Discussion prompt for the last five minutes: *"Your work is in a folder that syncs to the cloud. Explain why that is not the same thing as a backup."*

A tree, and directions through it

14 MIN

NAME

CLASS

DATE

A path is a set of directions

Folders form a tree. A **path** lists the folders you pass through, separated by slashes, ending at the file. Two files may share a name as long as they sit in different folders — the path is what makes each one unique.

the drive

FOLDER TREE

```
Documents/
  Grade10/
    DigitalTech/
      project.py
      notes.txt
    Science/
      report.docx
      notes.txt
  Photos/
    Camp/
      day1.jpg
      day1.png
```

Task 4.1 · Write the path

READ

#	FILE	FULL PATH
a	report.docx	
b	project.py	
c	day1.jpg	

d. Two files are called `notes.txt` . Why is that allowed?

e. Two files are called `day1` but they are not the same kind. What tells them apart? _____

f. You are in `Science` . What does `../DigitalTech/notes.txt` mean?

Names, types and copies

20 MIN

Task 4.2 · What the extension actually does

INVESTIGATE

FILE	WHAT OPENS IT	RISK	FILE	WHAT OPENS IT	RISK
essay.docx			project.py		
setup.exe			invoice.pdf		
grades.csv			photos.zip		
photo.jpg			report.pdf.scr		

a. One of these eight is dressed up as something it is not. Which, and how would you spot it?

b. You rename photo.jpg to photo.txt . What has changed inside the file? _____ What happens when you open it? _____

Task 4.3 · Is it actually backed up?

JUDGE

Your only copy of a year's work is on a laptop. For each arrangement, decide whether the work survives each disaster. Write Y or N.

THE ARRANGEMENT	LAPTOP STOLEN	FILE DELETED BY MISTAKE	RANSOMWARE ENCRYPTS EVERYTHING	VERDICT
A second copy in another folder on the laptop				
A folder that syncs to the cloud automatically				
An external drive that stays plugged in				
A USB drive in a drawer, last updated in March				

The rule professionals use is **3-2-1**: three copies, on two kinds of media, one of them somewhere else. Which arrangement above comes closest, and what is it still missing? _____

Task 4.4 · Deleted is not erased

PREDICT

When a file is deleted and the trash is emptied, the data is usually still on the drive. What the system removes is the *entry in the index* that says where it is. The space is marked as available and the contents sit there until something else is written over them.

#	QUESTION	YOUR ANSWER
a	Why can recovery software often get the file back?	
b	Why does waiting a week make recovery less likely?	
c	You sell a phone after deleting everything. What is the risk?	
d	What would genuinely destroy the contents?	

e. This behavior is deliberate, not a mistake. Suggest why deleting is designed to be this fast.

Task 4.5 · The last word

THINK

Your work sits in a folder that syncs to the cloud within seconds. Explain in one sentence why that is not a backup.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials

Task 4.1 · the paths

- Documents/Grade10/Science/report.docx
- Documents/Grade10/DigitalTech/project.py
- Documents/Photos/Camp/day1.jpg
- They sit in different folders, so their full paths differ. The name only has to be unique within its own folder.
- The extension — .jpg and .png — and they are also in different folders.
- .. means 'go up one level'. From Science, up to Grade10, then down into DigitalTech to find notes.txt.

Task 4.2 · extensions

.docx word processor · .exe a program, which runs whatever it likes — the highest risk here · .csv plain text table, opens in a spreadsheet · .jpg image viewer · .py text, but runs if handed to Python · .pdf document reader · .zip archive; the risk is what is inside it.

- report.pdf.scr. The real extension is the last one, and .scr is a program. It is designed to look like a PDF in a system that hides known extensions — which most do by default. Turning that setting off is genuinely useful advice.

- Nothing inside the file changes.** Every byte is identical. Only the label changed, so the system hands it to a text editor, which shows meaningless characters. Rename it back and the photo opens perfectly.

Task 4.3 · the four arrangements

ARRANGEMENT	STOLEN	DELETED	RANSOMWARE	VERDICT
Second folder, same laptop	N	Y	N	Not a backup at all
Cloud sync folder	Y	Usually — version history	Often N	Better, but a sync copies the damage as fast as it copies the work
External drive, always plugged in	N — it was in the same bag	Y	N — ransomware encrypts attached drives too	Half a backup
USB in a drawer, March	Y	Y	Y	A real backup, five months out of date

The USB comes closest — separate media, kept somewhere else, unaffected by anything that happens to the laptop. What it is missing is **currency**: a backup nobody updates is a backup of work you have since replaced. The best real answer combines the cloud folder with an offline copy updated on a schedule.

The sync argument is worth running properly. Sync is designed to make every copy identical, which is exactly the wrong behavior when the change you are copying is a deletion or an encryption.

Task 4.4 · deleted is not erased

- The contents are still on the drive; only the index entry pointing to them was removed. Recovery software scans the raw drive for data that looks like a file.
- A week of normal use writes new data into space marked as free, overwriting parts of the old file.
- The next owner can run recovery software and read messages, photos and documents you believed were gone.
- Overwriting the space with new data, a full factory reset that destroys the encryption key, or physically destroying the drive.
- Because removing one index entry takes a fraction of a second, while overwriting a 10 GB file would take minutes and wear out the drive. The design chose speed, and privacy is the cost.

Task 4.5 · the closing question

A backup is a copy taken at a moment in time and kept unchanged. A sync is a mirror that keeps everything identical — so the instant you delete, corrupt or encrypt the original, the mirror faithfully does the same thing.

Accept any answer reaching 'it copies the mistake too'. Strong students will mention version history, which is genuinely what turns a sync service into something backup-like.

Is It Finished?

Students turn six arguments into testable requirements, separate what a solution must *do* from how well it must do it, then evaluate a finished product using evidence rather than opinion.

You do not need to know this subject

There is no technology in this lesson. Every task is judgment about whether a statement can be proved, and the key gives a model for each one.

Photocopy these pages only

Pages 27-29

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

"Every argument at the end of a group project is really the same argument: nobody agreed at the start what finished meant. Today you find out how professionals stop that happening."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-12	Read the panel together	The word to repeat is: how would you prove it
12-26	Task 5.1 — rewrite the requirements	Judgment. Key has models
26-38	Task 5.2 — must do, or how well	Eight answers, all in the key
38-50	Task 5.3 — evaluate with evidence	The hardest task on the sheet
50-60	Tasks 5.4 and 5.5, collect sheets	Discussion prompt below

Questions students will ask

"Is that not just being picky?"

Ask them who wins the argument at the end if the requirement says 'fast'. Being picky at the start is how you avoid losing at the end.

"How can I evaluate something I have not built?"

You are evaluating the description on the page, and saying what evidence you would need. That is the actual skill.

"What if the client keeps changing their mind?"

That is Task 5.4, and the honest answer is that they will. The question is who pays for it.

If they finish early

Task 5.4 asks students to price three change requests. Strong students can then write the one sentence they would add to the original agreement to make that conversation easier.

If the computers are working

Students can find the published system requirements of any app or game and classify each line as something it must do or how well it must do it. Real ones are surprisingly badly written.

Curriculum focus · for the regular teacher's records

Defining problems; functional and non-functional requirements; developing design criteria; evaluating solutions against criteria using evidence. Discussion prompt for the last five minutes: *"Two people look at the same finished project. One says it works, one says it does not. Who decides, and what would settle it?"*

Provable, or arguable

14 MIN

NAME

CLASS

DATE

“It should be easy to use” is not a requirement

A requirement is only useful if two people who disagree can settle it by **testing**. If the only way to decide is to argue, you have written an opinion, not a requirement.

The test is simple: *how would somebody prove this was met?* If you cannot answer that in one sentence, rewrite it.

Vague

“The site should load quickly.”

Who decides what quickly means? On which connection? On what device?

Testable

“Any page loads in under 2 seconds on the school wifi, on a five-year-old laptop.”

Now it can be measured, and either it passed or it did not.

Task 5.1 · Make them provable

PROVE

#	AS WRITTEN	REWRITTEN SO IT CAN BE TESTED
a	The program should be fast	
b	It should work on most devices	
c	It should be secure	
d	The design should look professional	
e	It should handle a lot of users	
f	Errors should be handled properly	

Which of the six was hardest to make testable, and why? _____

What it does, and how well

Task 5.2 · Two kinds of requirement

CLASSIFY

A **functional** requirement says what the solution must *do*. A **non-functional** one says how well it must do it. Write F or N.

#	REQUIREMENT	F/N	#	REQUIREMENT	F/N
a	A student can book a locker		e	Staff can export the list as a spreadsheet	
b	Loads in under 2 seconds		f	Runs on a phone screen 5 inches wide	
c	Sends a confirmation email		g	Refuses two bookings for the same locker	
d	Works without a mouse		h	Stays available during the first week of the semester	

One of these eight would still let the system be completely useless even if it passed. Which, and why?

Task 5.3 · Evaluate with evidence

JUDGE

The locker booking system has been built. For each criterion, write the **evidence** you would need — not your opinion of it.

CRITERION	WHAT EVIDENCE WOULD SETTLE IT?
Students can book a locker without help	
It is faster than the paper system it replaced	
It never double-books	
Office staff can use it on day one	
It works for a student with a broken arm	

Two of these five can only be answered by watching real people. Which two? _____

Task 5.4 · Three change requests

DECIDE

The system is finished and agreed. The office now asks for three things. For each, decide whether it was already covered, and what it costs.

#	THE REQUEST	ALREADY AGREED?	WHAT IT COSTS
a	"Can the button be blue instead of green?"		
b	"Can it also handle bike cage bookings?"		
c	"It should have emailed parents. We assumed that."		

d. Request c is the dangerous kind. Write the one sentence you would have put in the original agreement to prevent it.

Task 5.5 · The last word

THINK

Two people look at the same finished project. One says it works and one says it does not. Explain in one or two sentences what would settle the argument, and who should have decided it — and when.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Task 5.1 · model rewrites

#	TESTABLE VERSION
a	Any screen responds within 2 seconds on the school wifi
b	Works in the current versions of the three browsers the school uses, on phone and laptop
c	Passwords are stored hashed; three failed logins lock the account for 10 minutes
d	Uses the school colors, one typeface, and text of at least 12pt throughout
e	Supports 200 people booking in the same 10-minute window without failing
f	Every invalid entry produces a message naming the field and how to fix it

c is the hard one and it is worth saying why: 'secure' is not one property, it is a long list of separate protections against separate threats. Rewriting it forces you to say which threats you actually care about — which is the entire job of security planning. Accept d as an alternative answer with reasoning, since 'professional' is taste rather than fact.

Task 5.2 · functional or not

F: a, c, e, g — things the system does.

N: b, d, f, h — how well, how fast, on what, for whom.

The dangerous one is b. A system can load in under two seconds and still book nothing at all. Non-functional requirements are worthless on their own, which is why a list made entirely of them describes a very fast, very reliable, very accessible product that does not work.

Accept d or h with the same reasoning.

Task 5.3 · evidence, not opinion

Book without help: watch ten students who have never seen it; count how many finish unaided.

Faster than paper: time both. The paper system took a measured average; so must this one.

Never double-books: a test where two people book the same locker at the same instant, run repeatedly. Absence of complaints is not evidence.

Office can use it on day one: ask a staff member to produce today's list with no training and record what happens.

Works with a broken arm: complete the whole task one-handed, and by keyboard only.

The two needing real people are the first and fourth. The others can be measured or tested by machine.

Task 5.4 · the change requests

a. Not agreed, but trivial. Minutes of work. Say yes; this is how goodwill is built.

b. Clearly new. A whole second booking system. It is a new project, or a new phase, with its own time and its own agreement.

c. The dangerous one. Nobody wrote it down, both sides genuinely believe they are right, and there is no evidence either way. This is the request that ends relationships.

d. Model sentence: 'Anything not written in this list is not included, and will be quoted separately.' Accept any version that makes the written list the only agreement. It sounds unfriendly and it prevents the only argument that cannot be settled.

Task 5.5 · the closing question

What settles it is the agreed criteria, written down before the work started, with the evidence each one requires. Without them, 'it works' means 'it does what I imagined' and the two people are describing different products.

The person who decides is the client or user, but only against criteria they agreed to in advance. Accept any answer that puts the deciding *before* the building.

Teaching a Machine

2 MINUTES

Students build a working spam filter by finding patterns in eight labeled examples, test it on new ones, then discover the two ways their perfectly accurate rule can be badly wrong.

You do not need to know this subject

The whole lesson is counting check marks in a table. Every score and every verdict is worked out in the key.

Photocopy these pages only

Pages 32-34

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. A calculator is not needed; the numbers are small.

WHAT TO SAY TO START

"Nobody writes the rules a modern system uses. They show it thousands of examples and it works out its own rule. Today you will do exactly that with eight examples and a pen, and then you will find out what goes wrong."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-12	Read the panel together	Rules written by hand versus patterns found in data
12-28	Task 6.1 — count the patterns and build the rule	Exact numbers. Key has all
28-40	Task 6.2 — test it on new mail	One result should annoy them. Good
40-52	Task 6.3 and 6.4 — bias and the accuracy trap	Best discussion in the volume
52-60	Task 6.5, collect sheets	Discussion prompt below

Questions students will ask

"Is this what AI actually does?"

It is a simplified version of a real technique. The difference is scale: millions of examples and hundreds of features, not eight and four.

"My rule got one wrong."

Then it is behaving like every real model ever built. The question is which mistake matters more, which is Task 6.2.

"Where did the rule come from?"

From the examples. Nobody wrote it. That sentence is the entire lesson and it is worth saying twice.

If they finish early

Task 6.4 is the accuracy trap and it defeats most adults. Strong students can then work out what question you should ask instead of 'how accurate is it?'

If the computers are working

If AI tools are permitted, students can try a simple text classifier or a teachable-machine style tool and deliberately train it on lopsided examples to reproduce Task 6.3 on screen.

Curriculum focus · for the regular teacher's records

Data and models; how systems learn patterns from training data; bias in data and its consequences; evaluating automated decisions.

Discussion prompt for the last five minutes: *"Nobody wrote the rule the machine is using. So who is responsible when the rule is unfair?"*

Finding the rule in the examples

16 MIN

NAME

CLASS

DATE

Two ways to make a machine decide

Written rules: a person decides what counts and types it in. You can read it, argue with it, and fix it.

Learned rules: the machine is shown examples already labeled with the right answer, and it finds whatever pattern separates them. Nobody chooses the pattern, and often nobody can say afterwards what it was.

Task 6.1 · Train the filter

ANALYZE

Eight emails have already been labeled by a human. Count how often each feature appears in each group.

EMAIL	LABEL	SHOUTING SUBJECT	2 OR MORE LINKS	UNKNOWN SENDER	SAYS 'FREE'
1	Spam	Y	Y	Y	Y
2	Spam	Y	Y	Y	N
3	Spam	N	Y	Y	Y
4	Spam	Y	N	N	Y
5	Not spam	N	N	N	N
6	Not spam	N	N	N	N
7	Not spam	N	Y	Y	N
8	Not spam	N	N	N	Y

COUNT	SHOUTING	2+ LINKS	UNKNOWN SENDER	'FREE'
How many of the 4 spam emails				
How many of the 4 real emails				

- Which single feature separates the two groups best? _____
- Score one point per feature present. What is the lowest score of any spam email? _____. The highest score of any real email? _____
- Write your rule: mark as spam if the score is at least _____

Testing what you built

18 MIN

Task 6.2 · Four emails it has never seen

TEST

#	THE EMAIL	SHOUTING	2+ LINKS	UNKNOWN 'FREE'	SCORE	YOUR RULE SAYS
W	Prize notification from a stranger	N	Y	Y	Y	
X	Your coach: TRAINING CANCELED	Y	N	N	N	
Y	Friend: FREE TICKETS!! with 2 links	Y	Y	N	Y	
Z	Bank asking you to confirm details, no links, polite	N	N	Y	N	

- Which email is wrongly marked as spam? ____ What does the user lose? _____
- Which dangerous email slips through? ____ Why can your features not catch it? _____
- Which of those two mistakes would you rather your filter made, and why?

Task 6.3 · It learned what you showed it

JUDGE

Each system below was trained on real data and works exactly as designed. Say what it will get wrong, and who it will get wrong.

#	WHAT IT WAS TRAINED ON	WHAT IT WILL GET WRONG
a	Face unlock, trained on photographs of adults aged 25–40	
b	An essay marker, trained only on essays that scored above 80	
c	A voice assistant, trained on speakers from one city	
d	A 'students at risk' model, trained on who was suspended last year	

Which of these four would be hardest to notice was going wrong? _____

Ninety-nine percent accurate

Task 6.4 · The accuracy trap

CALCULATE

A rare condition affects **1 person in 100**. A model is built to detect it. The model always answers “no”, whatever it is shown.

#	QUESTION	ANSWER
a	Out of 1,000 people, how many have the condition?	
b	How many of those does the model find?	
c	How many people does it label correctly overall?	
d	What is its accuracy as a percentage?	
e	Is it a good model?	

f. “99% accurate” would appear on the advertising. Write the question a buyer should ask instead.

Task 6.5 · The last word

THINK

Nobody wrote the rule your filter is using — you found it in the examples. If that rule turns out to be unfair, explain in one or two sentences who is responsible and why.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials

Task 6.1 · training the filter

In the 4 spam emails: SHOUTING 3, two-or-more links 3, unknown sender 3, 'free' 3.

In the 4 real emails: SHOUTING 0, links 1, unknown 1, 'free' 1.

a. SHOUTING — three of four spam, none of the real mail. The cleanest separator.

b. Every spam email scores **3 or more** (4, 3, 3, 3). No real email scores above **2** (0, 0, 2, 1).

c. Mark as spam if the score is **3 or more**. On the training data this rule is perfect — 8 out of 8. That perfection is exactly what Task 6.2 is about to spoil.

Task 6.2 · testing it

W score 3 → spam. Correct.

X score 1 → not spam. Correct.

Y score 3 → **spam** — wrong.

Z score 1 → **not spam** — wrong, and dangerously so.

a. Y. A real message from a friend is hidden in a folder nobody reads. That is a **false positive**, and users lose trust in a filter that does this.

b. Z is a phishing attempt written politely, with no links and no shouting. Every feature the model learned is about how spam *looks*, not what it *asks for*. A careful attacker simply avoids all four.

c. Either answer earns the credit with reasoning. Most email providers accept more spam in the inbox rather than risk losing real mail — the cost of the two mistakes is not equal.

Task 6.3 · what it learned instead

#	WHAT GOES WRONG	WHO PAYS
a	Fails more often on children and older faces, and on faces unlike those in the training photographs	Anyone outside the group that was photographed
b	Has never seen a weak essay, so it cannot recognize one. It grades everything as though it were good	Weaker students, who receive grades and feedback that mean nothing
c	Mishears accents it was never trained on and blames the speaker	Anyone who does not sound like the training data
d	Learns who <i>was suspended</i> , not who was at risk — so it reproduces last year's decisions, including any unfairness in them	Students from whichever groups were disciplined most often before

d is hardest to notice, because the model appears to be right: it flags students who then get into trouble, partly because being flagged brought extra attention. The prediction helps cause the outcome it predicted, and the numbers look like success the whole time.

The common thread worth drawing out: none of these systems is broken. Each one learned faithfully from what it was shown. The fault is in the choosing of the examples, and that choosing was done by people.

Task 6.4 · the accuracy trap

a. 10. **b.** None. **c.** 990. **d.** 99%. **e.** No — it is useless. It never detects the thing it was built to detect.

f. Good questions: 'of the people who have the condition, what fraction does it find?' and 'of the people it flags, what fraction really have it?' Accept any version that asks about the rare group separately.

This is why accuracy alone is close to meaningless whenever the thing being detected is rare — which is true of disease, fraud, and almost everything worth detecting.

Task 6.5 · the closing question

Responsibility sits with the people who chose the training examples, decided what counted as a feature, set the threshold, and put the system into use — because every one of those was a human decision that shaped the rule.

'The model decided' is not an answer. The model has no view about anything; it repeats patterns from the data it was given, at the scale somebody chose to deploy it.

Accept any answer that names a person or an organization rather than the software.

The Internet's Address Book

2 MINUTES

Students take a web address apart piece by piece, follow a name being turned into a number, and work out how somebody can type a perfectly correct address and still land on a fake site.

You do not need to know this subject

Everything is labeling and ordering. The completed URL diagram and the six lookup steps are in the key.

Photocopy these pages only

Pages 37-39

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

"Nothing on the internet knows what a website name is. Every name has to be turned into a number first, by somebody, and today you find out who that somebody is and what happens when they lie."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-14	Read the address apart together	Do the first two labels on the board
14-26	Task 7.1 — label the address	Exact answers. Grade as you circulate
26-38	Task 7.2 — order the lookup	Six steps, one correct order
38-48	Task 7.3 and 7.4 — one address, many devices; then the padlock	Key has both
48-55	Task 7.5, collect sheets	Discussion prompt below

Questions students will ask

"Is the domain the whole address?"

No. The domain is the part that was registered and paid for. Everything after the first single slash is chosen by whoever owns it.

"Does the padlock mean the site is safe?"

It means the connection is private. A criminal can have a padlock too. That distinction is Task 7.4.

"What is my IP address?"

Do not go looking. Keep the discussion to the school as a whole, which is what Task 7.3 is about.

If they finish early

Task 7.4 covers what an eavesdropper can and cannot see. Strong students can then work out why the *name* of the site you visit often leaks even when its contents do not.

If the computers are working

Students can run `nslookup` or `ping` on a well-known site and see the name become a number. Doing it twice from different networks sometimes returns different numbers, which is worth explaining.

Curriculum focus · for the regular teacher's records

Networks and protocols; addressing and the domain name system; secure transmission of data; evaluating threats to data in transit.

Discussion prompt for the last five minutes: *"You typed the address perfectly and still ended up on a fake site. Name every point in the journey where that could have happened."*



Names, numbers and the parts of an address

NAME _____

CLASS _____

DATE _____

Machines do not use names

Every device on the internet is found by a **number**, not a name. Names exist entirely for humans, and something has to translate one into the other before anything can be sent.

That translation service is the **domain name system**, and it is closer to a chain of address books than to a single one.

one address

URL

<https://library.school.edu:443/loans/search?title=nimona>

Task 7.1 · Take the address apart

LABEL

PART	WHICH PIECE OF THE ADDRESS IS IT?	WHAT IS IT FOR?
Protocol		
Subdomain		
Domain name		
Top-level domain		
Port		
Path		
Query		

- a. Which parts had to be bought and registered by the school? _____
- b. Which parts can be changed freely by whoever runs the server? _____
- c. A link reads `https://school.edu.login-secure.net/`. Which domain does it actually go to? _____

Turning a name into a number

18 MIN

Task 7.2 · Put the lookup in order

ORDER

Number these 1 to 6 to show what happens after you press Enter.

ORDER	STEP
	Your browser opens a connection to that number and asks for the page
	You type the address and press Enter
	The answer is stored for a while so the next lookup is instant
	Your device asks a name server: what number belongs to this name?
	Your device checks whether it already knows the number from earlier
	The name server replies with the number

a. Which step is skipped most of the time, and why does that make browsing faster?

b. The name server is down but the page still loads. How? _____

Task 7.3 · One address, nine hundred devices

THINK

The school has around 900 devices and **one** public address on the internet. Inside, each device has a private address that means nothing to the outside world.

#	QUESTION	YOUR ANSWER
a	What does a website see when you visit it from a school laptop?	
b	Can that website tell you apart from another student?	
c	Can the school tell you apart? How?	
d	One student breaks a website's rules. Who gets blocked?	

What the padlock does and does not mean

STRETCH

16 MIN

Task 7.4 · Who can see what

JUDGE

Somebody is watching the wifi between you and the internet. Check what they can see in each case.

WHAT THEY MIGHT SEE	OVER HTTP	OVER HTTPS
That you are online		
Which site you visited		
Which page on that site		
What you typed into the form		
The password you sent		

a. The padlock appears on a site called `school-portal-login.net`. What does the padlock actually prove?

b. What does it definitely *not* prove? _____

Task 7.5 · The last word

THINK

You typed the address perfectly and still landed on a fake site. Name two completely different points in the journey where that could have gone wrong.

- 1 _____
- 2 _____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

 Teacher initials _____

Task 7.1 · the parts of the address

PART	WHICH PIECE	WHAT IT IS FOR
Protocol	https	The rules for the conversation, and that it is encrypted
Subdomain	library	A section the school created; costs nothing and needs no registration
Domain name	school	The registered name
Top-level domain	.edu.au	The category and country; .edu.au has entry requirements
Port	443	Which door on the server; 443 is the normal one for https and is usually hidden
Path	/loans/search	Which page or function on that server
Query	?title=nimona	Extra information handed to the page

- a.** Only `school.edu` was registered. **b.** Subdomain, port, path and query are all free choices of whoever controls the domain.
c. It goes to `login-secure.net`. The real domain is always the part immediately before the first single slash, read right to left. Everything to the left of it is decoration, and this trick is used in phishing constantly.

Task 7.2 · the lookup order

- 1 You type the address and press Enter
 - 2 Your device checks whether it already knows the number
 - 3 It asks a name server
 - 4 The name server replies with the number
 - 5 The answer is stored for a while
 - 6 The browser connects to that number and asks for the page
- a.** Steps 3 and 4 are skipped whenever the answer is already remembered, which is most of the time. A lookup takes tens of milliseconds; a remembered answer takes none, and every page loads dozens of addresses.
- b.** The number is still in the cache from a previous visit. It will keep working until the stored answer expires.

Task 7.3 · one public address

- a.** The school's single public address — the same one it sees for every other student.
- b.** Not from the address alone. It can still tell devices apart using accounts, cookies and browser characteristics, which is worth mentioning.
- c.** Yes. The school's own equipment maps each private address to a device and usually to a login, and it keeps records of which device was given which address and when.
- d.** Potentially the whole school, because the site can only see the one address. This is a genuinely useful thing for students to know before they test it.

Task 7.4 · who sees what

	HTTP	HTTPS
You are online	Yes	Yes
Which site	Yes	Usually yes
Which page	Yes	No
What you typed	Yes	No
Your password	Yes	No

- a.** The padlock proves two things and no more: the connection is encrypted, and the certificate matches the name you actually connected to.
- b.** It proves nothing whatever about honesty, ownership or safety. Certificates are free and take minutes to obtain, so a criminal site has a padlock too. The padlock says 'nobody in between is listening', not 'the person at the other end is trustworthy'.

Task 7.5 · the closing question

Any two of: the name server gave a false answer, deliberately or because it was compromised · the device's stored answer was poisoned earlier · somebody controlling the network redirected the traffic · the address was correct but the domain was a look-alike, differing by one character · the link you followed displayed one address and led to another.

The useful conclusion: correctness of what you typed is only one link in a chain, and you do not control the rest of it.

How a Search Engine Ranks

2 MINUTES

Students build a real index from four tiny pages, answer searches using only that index, then rank five pages by counting links and discover how easily counting can be cheated.

You do not need to know this subject

Every answer is a list of page names you can read straight off the tables. The completed index and the rankings are in the key.

Photocopy these pages only

Pages 42-44

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

"A search engine does not look at the web when you search. It looked months ago and wrote everything down. Today you build the thing it wrote down, and then you find out how people cheat it."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-14	Read the crawl-index-rank panel	The dictionary comparison does the work
14-30	Task 8.1 — build the index and search it	Exact answers. Key has the full index
30-44	Task 8.2 — rank by links, then break it	The link ring is the good part
44-54	Task 8.3 and 8.4 — relevance, and better questions	Judgment. Key has models
54-60	Task 8.5, collect sheets	Discussion prompt below

Questions students will ask

"Do I write every word in the index?"

Yes, except the very common ones. Task 8.1 tells them which to leave out and why.

"Why not just search the pages when I ask?"

There are hundreds of billions of pages. Reading them at the moment you search would take weeks; reading an index takes milliseconds.

"Is this how it really works?"

The index part, yes, almost exactly. The ranking part uses hundreds of signals, but counting links was the original idea and it is still in there.

If they finish early

Task 8.4 asks students to rewrite four hopeless searches. Strong students can then explain why a search engine that showed the most *truthful* page first would be extremely difficult to build.

If the computers are working

Students can compare the same query with and without quotation marks, and with a minus sign in front of a word. The difference in results is immediate and makes Task 8.4 concrete.

Curriculum focus · for the regular teacher's records

How data is indexed and retrieved; algorithms for searching and ranking; evaluating the reliability and authority of online sources. Discussion prompt for the last five minutes: *"The first result is the one almost everybody clicks. What is it actually the best at being?"*

Crawl, index, rank

16 MIN

NAME

CLASS

DATE

It looked before you asked

A search engine sends out programs that follow links and read pages — that is **crawling**. What it reads is turned inside out into an **index**: a list of words, and for each word, every page that contained it. Searching reads the index, never the web.

Ranking is the separate question of which of the matching pages to show first, and that is where the money and the arguments are.

Task 8.1 · Build the index

DO

Four very short pages. Ignore the words **for**, **and** and **the** — they appear everywhere and separate nothing.

PAGE	ITS ENTIRE CONTENTS
P1	school cafeteria menu prices for hot food
P2	hot air balloon festival tickets and prices
P3	cafeteria volunteer schedule for the fall semester
P4	balloon animals for the school fair

WORD	APPEARS ON	WORD	APPEARS ON
school		balloon	
cafeteria		tickets	
menu		volunteer	
prices		animals	
hot		fair	

Now search. A page matches only if it contains **every** word.

SEARCH	PAGES RETURNED
hot balloon	
school cafeteria	
prices	
cafeteria balloon	

Which one goes first?

18 MIN

Task 8.2 · Rank by links

CALCULATE

A link from one page to another is a kind of vote. Here is who links to whom.

PAGE	LINKS OUT TO
A	C
B	C
C	E
D	A, C
E	C

PAGE	LINKS IN	RANK
A		
B		
C		
D		
E		

Then this happens

Three brand new pages, **S1**, **S2** and **S3**, appear. Each links only to **B** and to each other. Nothing anywhere links to them.

a. B now has how many links in? _____

b. Is B now a better page? _____

c. What extra rule would stop this working?

d. C and E link to each other. What does that do to a system that only counts, and why is it a problem?

Task 8.3 · Relevance is not authority

JUDGE

Somebody searches **is paracetamol safe for a 6 year old** . Put these four results in the order you would show them, and say why.

RESULT	ORDER	WHY
A blog post repeating the phrase 32 times		
A government health service page		
A forum thread with 200 replies		
A shop selling paracetamol		

Which result best matches the *words* of the search, and why is that not enough?

Task 8.4 · Fix the search

EXTENSION

#	WHAT THEY TYPED	WHY IT FAILS	A BETTER SEARCH
a	jaguar — wanting the animal		
b	how do I do the thing in python		
c	ap chemistry course description — wanting the official one		
d	best phone		

e. Search d is the hardest of the four to fix. Explain why. _____

Task 8.5 · The last word

THINK

Almost everybody clicks the first result. Explain in one or two sentences what the first result is actually the best at being — and what it is not.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

How a Search Engine Ranks

ANSWERS

Task 8.1 · the index

school → P1, P4 · cafeteria → P1, P3 · menu → P1 · prices → P1, P2 · hot → P1, P2 · balloon → P2, P4 · tickets → P2 · volunteer → P3 · animals → P4 · fair → P4.

Searches: hot balloon → **P2** · school cafeteria → **P1** · prices → **P1 and P2** · cafeteria balloon → **nothing**.

The last one matters: no page contains both words, so the honest answer is no results. Ask the class what a search engine does instead, and why showing something is commercially better than showing nothing.

Task 8.2 · counting links

Links in: A = 1 (from D) · B = 0 · **C = 4** (A, B, D, E) · D = 0 · E = 1 (from C).

Rank: C first, then A and E equal, then B and D.

a. Three. **b.** No — nothing about B changed. Three worthless pages voted for it.

c. Weight each link by how important the *linking* page is. A vote from a page nobody links to is worth almost nothing. That is the core of the original PageRank idea and it is worth naming once a student has reinvented it.

d. Mutual links inflate both pages at once, and a group doing it deliberately is called a link farm. Counting alone cannot tell an honest recommendation from an arrangement.

Task 8.3 · relevance and authority

ORDER	RESULT	WHY
1	Government health service	Accountable, reviewed, and correct if wrong is a serious matter for them
2	Forum thread	Real experience, sometimes useful, entirely unverified
3	Blog post	Written to rank, not to inform
4	Shop	Has a financial interest in one answer

The blog matches the words best — it contains the exact phrase far more often than any other page. That is precisely why word-matching alone stopped working in the 1990s: it rewards whoever repeats the phrase most, which is a thing anybody can do on purpose.

Accept a different order if the reasoning names authority, accountability or conflict of interest. For a medical question in particular, it is worth saying aloud that no search result replaces a pharmacist or doctor.

Task 8.4 · better searches

- The word has three common meanings. Add a word only one meaning uses: jaguar habitat rainforest .
- Nothing in it is distinctive. Name the actual task and the error: python list index out of range .
- Thousands of tutoring sites outrank the source. Restrict it: ap chemistry course description site:collegeboard.org .
- Add the constraints that matter to you: phone under \$600 best battery 2026 review .
- Because 'best' is not a property of a phone; it depends on budget, use and taste. Worse, the entire results page for that phrase is written by people paid when you buy something. No wording fixes a question that has no factual answer.

Task 8.5 · the closing question

The first result is the best at satisfying the ranking system: matching the words, being linked to, loading quickly, and being clicked by people who did not immediately come back.

It is not the truest, the most careful, or the most complete. Nothing in the ranking measures whether a page is correct, because no algorithm can check that — and a page written specifically to rank first will always beat a better page that was not.

Accept any answer that separates 'most popular and best optimized' from 'most correct'.

Machines That Sense and Act

2 MINUTES

Students write the decision table for a line-following robot, plan a route through a maze, then work out why a light sensor at dusk makes a lamp flicker on and off all evening.

You do not need to know this subject

No robot and no code. Every answer is a direction, a number or a short rule, and all of them are in the key.

Photocopy these pages only

Pages 47-49

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

"A robot has no idea where it is. It knows two numbers from two sensors, and it has to decide what to do with them thirty times a second. Today you write those decisions."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-12	Read the control loop panel	Sense, decide, act, repeat
12-26	Task 9.1 — the decision table	Four rows. Two are obvious, two are not
26-38	Task 9.2 — the maze	Count the moves. Key has the route
38-50	Task 9.3 and 9.4 — thresholds and stopping	The flickering lamp is the good one
50-55	Task 9.5, collect sheets	Discussion prompt below

Questions students will ask

"Which way does it turn if both sensors see the line?"

That is the interesting row and there is no single right answer. Ask them to justify whichever they choose.

"Can it just remember the map?"

Only if somebody gave it one. Task 9.2b assumes it has no map at all, which is the harder and more realistic case.

"What is a threshold?"

The number where a decision flips from one answer to the other. Choosing it badly is Task 9.3.

If they finish early

Task 9.4 is about stopping safely. Strong students can then explain why a robot that has lost its line should stop rather than guess, and why that is not always the safe choice for a vehicle.

If the computers are working

Any robot simulator or block-coding environment with virtual sensors reproduces Task 9.1 directly. The paper table first, or the trial-and-error replaces the thinking.

Curriculum focus · for the regular teacher's records

Control systems and algorithms; sensors, thresholds and calibration; iteration and branching in control; safety in automated systems.

Discussion prompt for the last five minutes: *"A robot did exactly what it was programmed to do and something went badly wrong. Why is "it followed the program" not a defense?"*

Sense, decide, act, repeat

14 MIN

NAME _____

CLASS _____

DATE _____

The control loop

A controlled machine runs the same four steps forever: **read the sensors, decide, act**, then read them again. Nothing is remembered between passes unless you deliberately store it.

A line-following robot has two light sensors underneath it, one on each side of a black line. Each one reports either **dark** (it is over the line) or **light** (it is over the floor).

Task 9.1 · Write the decision table

DECIDE

#	LEFT SENSOR	RIGHT SENSOR	WHAT THAT MEANS	WHAT THE ROBOT SHOULD DO
1	Dark	Dark		
2	Dark	Light		
3	Light	Dark		
4	Light	Light		

a. Which row happens when the robot is perfectly on course? _____

b. Row 4 has two sensible answers. Give both, and say when each would be better.

c. The loop runs 30 times a second. What would you see if it ran only twice a second?

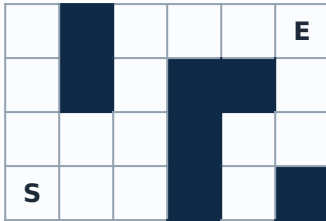
Finding a way through

18 MIN

Task 9.2 · The maze

PLAN

The robot starts at **S** facing right and must reach **E**. Shaded squares are walls. It can only move forward one square at a time or turn 90 degrees on the spot.



Write the instructions

Use FORWARD n , TURN LEFT and TURN RIGHT .

a. Forward moves in total: _____

b. Turns in total: _____

c. Now the robot has **no map** — only a sensor that tells it whether there is a wall on its right. Write a rule, repeated forever, that would still get it out of a maze like this one.

Task 9.3 · Choosing the threshold

CALCULATE

A light sensor reports a number from 0 to 1023. In the lit room it reads about **800**. With the lights off it reads about **120**. The lamp should switch on when it is dark.

#	QUESTION	ANSWER
a	Choose a threshold and write the rule	
b	Why is a threshold of 790 a bad choice?	
c	At dusk the reading drifts around your threshold. What does the lamp do?	
d	Fix it using two numbers instead of one	

Task 9.4 · The stop button

JUDGE

Every serious machine has an emergency stop. Where it sits in the system matters more than what it looks like.

#	DESIGN	WOULD IT ALWAYS WORK?	WHY
a	A button the program checks once every loop		
b	A switch that cuts power to the motors directly		
c	A command sent over wifi from a laptop		

d. Design a is the most common in school projects. Name the exact situation where it fails.

e. A robot loses the line completely. Should it stop or keep searching? Now answer the same question for a car that loses its lane markings.

Task 9.5 · The last word

THINK

“The robot did exactly what I programmed.” Explain in one or two sentences why that sentence is true and is still not a defense.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

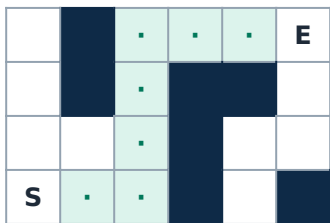
Teacher initials

Task 9.1 · the decision table

#	LEFT	RIGHT	MEANING	ACTION
1	Dark	Dark	Both over black — a junction, a thick line, or a finish marker	Go straight, or stop if this marks the end
2	Dark	Light	Drifting to the right of the line	Turn left
3	Light	Dark	Drifting to the left of the line	Turn right
4	Light	Light	Both over the floor — the line has been lost	Stop, or turn back toward where it was last seen

- a.** Neither 2 nor 3 is 'on course' exactly — a two-sensor robot straddles the line and corrects constantly. Accept row 1 for a thick line, and accept 'it is never perfectly on course, it wobbles', which is the better answer.
- b.** Stopping is safe and predictable. Searching (turn back the way it last corrected) recovers from a gap in the tape. Stopping is better where a lost robot could do damage; searching is better on a course with deliberate gaps.
- c.** At two decisions a second the robot overshoots wildly between corrections and zigzags off the line entirely. How often the loop runs is as much a part of the design as the rules inside it.

Task 9.2 · the maze



Route: FORWARD 2 (to the third column), TURN LEFT, FORWARD 3 (to the top row), TURN RIGHT, FORWARD 3 to E.

- a.** 8 forward moves. **b.** 2 turns.
- c.** The wall-follower: *if there is no wall on your right, turn right and go forward; otherwise if you can go forward, go forward; otherwise turn left.* Repeat forever. It is not the shortest route and it will not solve every maze, but it needs no map and no memory at all — which is why it is the first algorithm every robotics club meets.

Task 9.4 · stopping

- a. Not always.** If the program crashes, freezes or is stuck in a loop, it never checks the button again and the motors keep running.
- b. Yes.** Cutting power needs no working software, so it is the only design that survives the failure you most need it for. Real emergency stops are wired this way by law.
- c. No.** It depends on wifi, on the laptop, and on the program — three things that can each fail independently.
- d.** The exact failure is: the software has stopped responding but the motors are still powered. That is precisely the moment somebody presses the button.
- e.** A line robot should stop — nothing bad happens when a stopped robot sits still. A car cannot stop instantly in traffic, so it must hand control back to the driver, keep its lane by other means, and slow safely. Same question, opposite answer, because the cost of stopping is different.

Task 9.3 · the threshold

- a.** Anything near the middle: *if the reading is below 450, switch the lamp on.* Accept 300–600 with reasoning.
- b.** 790 sits almost on top of the lit-room reading, so ordinary changes — a cloud, somebody walking past — cross it constantly.
- c.** The lamp switches on, which does not change the outside light but does change what the sensor sees if it is nearby, and the reading wanders back and forth across the line. The lamp flickers on and off, possibly for an hour.
- d.** Use two thresholds: switch **on** below 400, switch **off** only above 600. Between the two, do nothing. The gap is called **hysteresis** and every thermostat, streetlight and automatic door uses it.

Task 9.5 · the closing question

It is true and irrelevant. The machine has no judgment, so the only judgment in the system was the programmer's — including everything they failed to consider. A rule that behaves perfectly in the situations you imagined is not a rule that behaves safely in the ones you did not.

Accept any answer reaching 'the program is what you decided, so its mistakes are your decisions'.

Whose Work Is It?

2 MINUTES

Students decide whether eight everyday uses are allowed, match five licenses to what they actually permit, and write a correct attribution. Discussion-heavy, and immediately useful in every other subject.

You do not need to know this subject

This is reading and judgment. The key gives the reasoning for every scenario and flags the two that genuinely have no settled answer.

Photocopy these pages only

Pages 52-54

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

"Everything anybody makes is automatically owned by them from the moment it exists. That includes everything you have ever made, and everything you have ever used without asking."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-12	Read the panel together	Automatic, and a license is permission
12-28	Task 10.1 — can I use it	Expect argument. Argument is the lesson
28-38	Task 10.2 — match the licenses	Five answers, all in the key
38-48	Task 10.3 — write the attribution	Four elements. Key has the model
48-55	Tasks 10.4 and 10.5, collect sheets	Discussion prompt below

Questions students will ask

"But everyone does it."

True, and not an answer. Ask whether the use is allowed, then separately whether anyone would ever notice. They are different questions.

"Do I have to register a copyright?"

No. It exists the moment the work does. Registration systems exist in some countries but protection here is automatic.

"Is it fine if I credit them?"

Crediting is not permission. It is required by some licenses and it satisfies none on its own.

If they finish early

Task 10.4 asks where inspiration ends and copying begins. Strong students can then argue the case for the shortest possible copyright term and the case for the longest, which is a live legal argument.

If the computers are working

Students can use an image search filtered by usage rights and see how much smaller the result set becomes. Finding the license terms of a font is a surprisingly good exercise.

Curriculum focus · for the regular teacher's records

Legal and ethical responsibilities in the use of digital content; intellectual property, licensing, attribution and the rights of creators. Discussion prompt for the last five minutes: *"You made it in class, on a school computer, in school time. Who owns it — and how would you find out?"*

Automatic, and yours

14 MIN

NAME

CLASS

DATE

Nobody has to claim it

Copyright is automatic. The moment you write, draw, photograph, record or code something original, you own it. No symbol, no registration, no notice required.

A **license** is permission granted in advance to everybody, or to somebody in particular. It does not give away ownership; it says what other people may do without asking first.

Attribution is naming the creator. Some licenses require it. None of them are satisfied by it alone.

Task 10.1 · Can I use it?

JUDGE

Mark each **allowed**, **not allowed** or **it depends**, and give your reason.

#	WHAT THE STUDENT DID	VERDICT	REASON
a	Used an image from a search engine in a class assignment		
b	Used the same image in a video on a channel that earns money		
c	Used a CC BY photo and named the photographer		
d	Used a song as the soundtrack of a school assembly video		
e	Copied code from a public repository with an MIT license		
f	Scanned two pages of a textbook for their own study		
g	Reposted a meme built from a film still		
h	Downloaded a font marked 'free for personal use' for a school poster		

Which two of these eight genuinely have no clear answer? _____

Task 10.2 · Match the license

CLASSIFY

LICENSE	MAY I USE IT?	MUST I CREDIT?	MAY I SELL WHAT I MAKE?	MAY I CHANGE IT?
All rights reserved				
CC BY				
CC BY-NC				
CC0 / public domain				
MIT (software)				

- a. Which one asks for the least and gives the most? _____
- b. Your school YouTube channel runs advertisements. Which license in the list has just become a problem, and why? _____

Task 10.3 · Write the attribution

DO

A good attribution names four things: the **title**, the **creator**, the **license**, and where you **found** it.

WHAT YOU KNOW	DETAIL
Title	<i>Morning at Cedar Point</i>
Creator	R. Whitcombe
License	CC BY 4.0
Found at	openphotos.example/nepean

Write the attribution line as it would appear under the photo in your slide:

- a. You cropped the photo. What must you add? _____
- b. Why is 'image from the internet' worse than no caption at all? _____

Task 10.4 · Three hard cases

DECIDE

#	THE CASE	COPYING, OR NOT? WHY?
a	A game with the same rules as a famous one, new art, new name	
b	A song with a four-second sample of another song, slowed down	
c	A program written by reading somebody else's code, then typing your own	

d. What is the difference between copying an *idea* and copying an *expression* of it? Which one does copyright cover?

Task 10.5 · The last word

THINK

You made it in class, on a school computer, in school time. Explain in one or two sentences who you think owns it, and say exactly how you would find out for certain.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Task 10.1 · can I use it

#	VERDICT	REASONING
a	Usually allowed	Most countries allow limited copying for study or classroom use. It is narrower than students assume and it does not survive publication
b	Not allowed	Earning money removes the education argument entirely. This is the single clearest line on the page
c	Allowed	CC BY exists precisely for this, and the credit is the condition being met
d	It depends	Schools hold blanket music licenses that cover a great deal. Ask; do not assume either way
e	Allowed	MIT permits use, change and sale, provided the license notice is kept with the code. Deleting the notice is the breach students actually commit
f	Usually allowed	Copying a small portion for personal study is generally permitted
g	It depends	Widely tolerated, rarely licensed. The rights holder could object and occasionally does
h	Not allowed	'Personal use' excludes an organization, and a school is one. The license says what it says

The two with no clear answer are d and g. Both depend on licenses held by somebody else or on whether a rights holder chooses to act. Students who say 'it depends' and explain what it depends on have given the better answer than a confident yes or no.

Task 10.2 · the licenses

LICENSE	USE	CREDIT	SELL	CHANGE
All rights reserved	Ask first	—	No	No
CC BY	Yes	Required	Yes	Yes
CC BY-NC	Yes	Required	No	Yes
CC0	Yes	Not required	Yes	Yes
MIT	Yes	Keep the notice	Yes	Yes

a. CC0 — the creator has given up everything they can give up. Crediting is still good manners.

b. CC BY-NC. The moment the channel earns money the use is commercial, and NC forbids it. This catches schools and clubs constantly.

Task 10.3 · the attribution

Model: *Morning at Cedar Point* by R. Whitcombe, licensed under CC BY 4.0, from openphotos.example/nepean

a. A note that the work was modified — 'cropped from the original'. CC licenses require changes to be indicated so nobody blames the creator for your edit.

b. Because it looks like an attribution and is not one. Nobody can find the source, check the license or credit the right person, and the reader is misled into thinking the question was handled. No caption at least leaves the gap visible.

Task 10.4 · the hard cases

a. Generally not copying. Game rules and mechanics are ideas; art, names, characters and text are expression. Copying the mechanics is legal and is how entire genres exist.

b. Copying. Length does not rescue it. Using a recording at all requires clearance, and altering it does not make it yours — this has been settled in court repeatedly.

c. Depends on what you carried across. Learning a technique is fine. Reproducing the same structure line by line from memory is still copying, even though you typed every character.

d. Copyright protects the **expression** — the particular words, images, sounds or code — never the underlying idea. Anyone may write another story about a wizarding school; nobody may reprint the sentences.

Task 10.5 · the closing question

The honest answer is that it varies. Many schools and systems have a policy saying student work belongs to the student; some employment and enrollment agreements say otherwise; and equipment ownership does not by itself transfer copyright.

The part worth marking is **how you would find out:** read the school's ICT or intellectual property policy, ask the person who runs it, and check anything you signed at enrollment. 'Look it up in the policy' is a better answer than any confident guess.

This matters more than it sounds. Students publish, sell and license their own work far more often than they expect to.

Photocopy this page or fill it in directly. Ninety seconds of your time saves the regular teacher a great deal of guesswork.

SUBSTITUTE TEACHER

CLASS

DATE

PERIOD

Lesson used

CHECK ONE

☐ 01 · Lists, Indexes and Off-by-One☐ 06 · Teaching a Machine☐ 02 · True, False and In Between☐ 07 · The Internet's Address Book☐ 03 · Inside the Box☐ 08 · How a Search Engine Ranks☐ 04 · Where Did the File Go?☐ 09 · Machines That Sense and Act☐ 05 · Is It Finished?☐ 10 · Whose Work Is It?**How far did the class get?**

- ☐ Everyone finished the main tasks
- ☐ Most finished; some are partway
- ☐ We got through about half
- ☐ We did not get far — see notes

Sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Graded against the answer key

Anything the regular teacher should know

NOTES

Students to mention · for any reason, good or otherwise

NAMES

Thank you

Walking into an unfamiliar subject is not easy. Leaving a clear note behind is the part that makes the next teacher's day work, and it is genuinely appreciated.

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Curriculum alignment

CSTA K-12 Computer Science Standards, levels 2 and 3A. Across the ten lessons this volume touches:

- Data structures: lists, indexing and iteration
- Boolean logic, truth tables and branching conditions
- Hardware components and the fetch-execute cycle
- File management, backup and secure deletion
- Requirements, design criteria and evaluation with evidence
- Machine learning, training data and bias
- Network addressing, DNS and secure transmission
- Searching, indexing and ranking algorithms
- Control systems, sensors, thresholds and safety
- Intellectual property, licensing and attribution

Individual lesson teacher pages name the specific focus for that lesson.

The rest of the series

Volume 1 · Foundations

Algorithms, binary, trace tables, flowcharts, cyber attacks, networks, sorting, debugging, AI and digital footprints.

Volume 2 · Data, Systems and Design

Data cleaning, misleading charts, databases, input-process-output, compression, encryption, interface design, functions, project planning and the cost of free apps.

Volume 4 · Media, Evidence and Impact

Images and sound as data, stacks and queues, digital forensics, version control, network design, accessibility, algorithmic decisions, IoT security and automation.

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Physical Computing with the Raspberry Pi 5 · AI Literacy for Secondary Students · and more.

Found an error, or want a lesson on something else?

Feedback is genuinely read. If a lesson did not land with your class, or you need a sub plan on a topic that is not here, get in touch through digitalvector.com.au and it will be considered for the next update.

EMERGENCY SUB PLANS PACK

Digital Technologies · Grades 7-10

Volume Three · ten lessons · no preparation · no computers required